

A complete fixed-target cofactor classification and original Erdős–Straus channel returns

The Clankers

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Abstract

We classify all divisors of j^2 in a fixed positive class t modulo $4j - 1 > t$. A canonical divisor and an unbounded companion are separated from a finite exceptional set with sharp grade bound $4j - 1 \leq t^2 - 3t + 1$. The bound is realized by original exterior states at infinitely many hard primes. Applied to original negative shape coordinates, the classification gives complete same-grade middle returns and forced availability on entire nonsquare shape lines. It also proves that each fixed alpha coefficient $t \geq 4$ has infinitely many original hard-prime examples with no same-grade middle return. This is not a universal ES proof.

Theorem 1 (Complete capacity and sharp boundary). *For $h = 4j - 1 > t \geq 1$, define $\mathcal{A}_t(j) = \{W \mid j^2 : W \equiv t \pmod{h}\}$. Its disjoint, uniquely labelled words are*

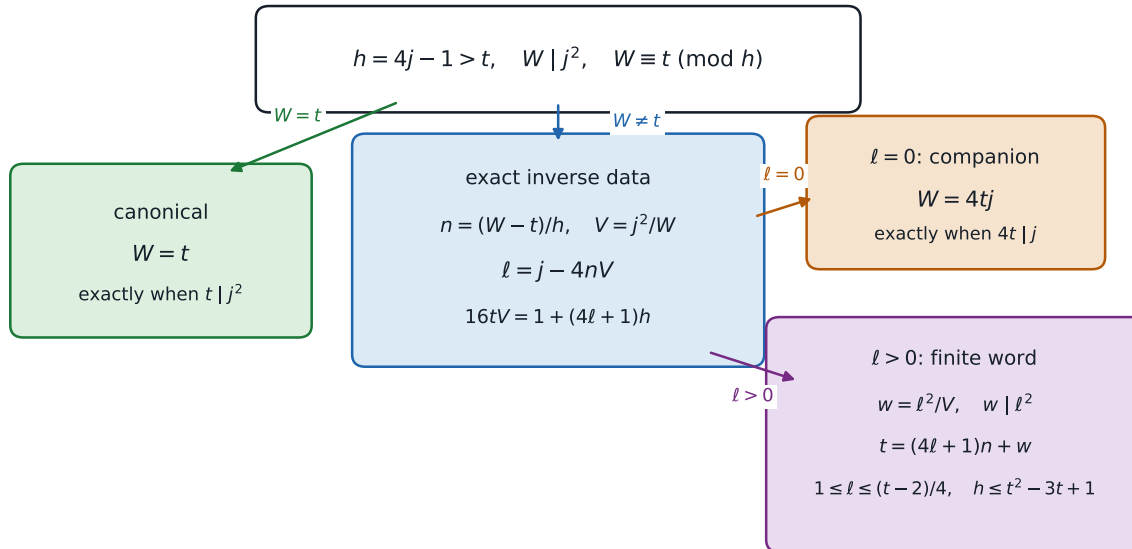
$$W = t \quad (t \mid j^2), \quad W = 4tj \quad (4t \mid j), \quad (1)$$

and the finite words

$$1 \leq \ell \leq \lfloor (t-2)/4 \rfloor, \quad w \mid \ell^2, \quad n = (t-w)/(4\ell+1) \in \mathbb{Z}_{>0}, \\ V = \ell^2/w, \quad j = \ell + 4nV, \quad h > t, \quad W = j^2/V = t + nh. \quad (2)$$

Every finite word satisfies $h \leq t^2 - 3t + 1$. Equality holds exactly at $w = n = 1$, $t = 4\ell + 2$. In particular, a nonempty box with $t \nmid j^2$ is in this finite grade range. Beyond it the exact count is $\mathbf{1}_{t \mid j^2} + \mathbf{1}_{4t \mid j}$.

Exact inverse classification of a fixed-target cofactor word



The three outputs are disjoint and the displayed formulas recover every label. The quadratic bound applies only to the finite branch.

Figure 1: Exact inverse classification in Theorem 1. The canonical, companion and finite branches are disjoint, and every label is recovered from the original divisor W .

Proof. For $W \neq t$, write $W = t + nh$, $n \geq 1$, and $V = j^2/W$. The identity $16WV = (h+1)^2$ gives $16tV = 1 + kh$, where $k \geq 1$ and $k \equiv 1 \pmod{4}$. Put $k = 4\ell + 1$. Subtraction gives

$$j = \ell + 4nV, \quad t = (4\ell + 1)n + \ell^2/V.$$

At $\ell = 0$, $n = t$ and $j = 4tV$, exactly the second word of (1). At $\ell > 0$, $w = \ell^2/V$ is a positive integer and $t \geq 4\ell + 2$. All steps reverse, with inverse $n = (W - t)/h$, $V = j^2/W$, $\ell = j - 4nV$. Thus the list has no omissions or duplicate labels. Finally

$$h + 1 = 4\ell + \frac{16\ell^2(t - w)}{(4\ell + 1)w} \leq 4\ell + \frac{16\ell^2(t - 1)}{4\ell + 1} \leq t^2 - 3t + 2.$$

The last expression increases strictly in ℓ , whose maximum is $(t - 2)/4$. Equality conditions give $w = n = 1$. The companion always coexists with the canonical word, so it cannot repair its absence. $\square$

The sharp family is

$$t = 4\ell + 2, \quad j = \ell(4\ell + 1), \quad h = t^2 - 3t + 1, \quad W = (4\ell + 1)^2, \quad V = \ell^2. \quad (3)$$

Here $t \nmid j^2$, because $2\ell + 1 > 1$ is coprime to j . Finite words can also coexist with the canonical one: $\mathcal{A}_{289}(1224) = \{289, 5184\}$. The companion in (1) is genuinely unbounded and must not be folded into the finite exceptional claim.

Original arithmetic and inverse. For prime $p \equiv 1 \pmod{4}$, an original exterior state has

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad R \mid 4u + 1.$$

Normalize by $g = \gcd(a, u)$, $(h, r, s) = (g^2/u, u/g, a/g)$ and $\kappa = (pr + s)/R$. The original ordered return is $(a, h s \kappa, p h r \kappa)$; its inverse is $u = a^2/(Ry - pa)$. Keep

$$v = h s^2, \quad D = (4u + 1)/R, \quad \alpha = v - R, \quad \beta = v - D.$$

The original identities yield $h \mid \alpha\beta - 1$, $p \equiv \alpha \pmod{h}$, $p\beta \equiv 1 \pmod{h}$. Reciprocity on the actual factors gives $(h/p) = -1$.

If $\alpha = -4t$ or $\beta = -4t$, then $h \equiv 3 \pmod{4}$, s is odd, $\gcd(h, t) = 1$ and $(t/h) = 1$. Set $j = (h + 1)/4$. The entire middle cofactor- h gate is exactly

$$W \in \mathcal{A}_t(j), \quad U = W(\alpha), \quad U = j^2/W(\beta).$$

For every word it returns

$$R_M = (p + 4U)/h, \quad a_M = (p + R_M)/4, \quad g_M = \gcd(j, U), \\ h_M = g_M^2/U, \quad r_M = U/g_M, \quad \lambda_M = j/g_M, \quad s_M = R_M \lambda_M - r_M, \quad (4)$$

with denominators $(a_M, p h_M s_M \lambda_M, p h_M r_M \lambda_M)$. Indeed $U \mid j^2$, $h \mid p + 4U$, and $4U \leq (h + 1)^2/4 < (h - 1)p$ since $p > 2h$. Thus $0 < R_M < p$ and $R_M \equiv 3 \pmod{4}$. Prime valuations give the integral normalization; $a_M = R_M j - U = h_M r_M s_M > 0$, and a common divisor of r_M, R_M would divide p , although $r_M \leq j < p$. This proves the original coprimality and both range bounds. Complementing U exchanges the retained middle orientation.

These are classical Type-II coordinates [1, §2]; the contribution is the complete capacity input and its original source supply. In (2), putting $g_0 = \gcd(\ell, V)$ gives $H_0 = g_0^2/V$, $r_0 = V/g_0$, $\lambda_0 = \ell/g_0$. The returned (h_M, r_M, λ_M) equals $(H_0, \lambda_0 + 4nr_0, r_0)$ or $(H_0, r_0, \lambda_0 + 4nr_0)$. Thus $h_M \leq \ell < t/4$ on the finite branch.

Theorem 2 (Automatic entire shape lines). *At a prime $p \equiv 1 \pmod{24}$, the original source forces $t \mid j^2$ on*

$$\alpha = -4t, \quad t \in \{1, 2, 3\}; \quad \beta = -4t, \quad t \mid 36.$$

Every such source therefore has the explicit middle return (4), without any bound on its other shape coordinate.

Proof. For alpha, $(2/h) = 1$ with $h \equiv 3 \pmod{4}$ forces $h \equiv 7 \pmod{8}$; $(3/h) = 1$ forces $h \equiv 11 \pmod{12}$. These give $2 \mid j$ or $3 \mid j$. For beta, $D = h s^2 + 4t$, $r + \kappa = sD$ odd and $pD + 1 = 4hr\kappa$ give $D \equiv 7 \pmod{8}$ and $h \equiv 7 - 4t \pmod{8}$. If $2 \mid t$, then $2 \mid j$. If $3 \mid t$, all equalities in the remainder of this sentence are modulo three: the case $h = 0$ is impossible; at $h = 1$, $s = 0$ would require $1 = -r^2$; $s = \pm 1$ would require $2 = r(s - r)$, also impossible. Hence $h \equiv 2 \pmod{3}$ and $3 \mid j$. Since each exponent in $t \mid 36$ is at most two, this proves the full divisor budget. $\square$

Sharpness in the original prime domain. For every $t = 6 + 420z$, put $h = t^2 - 3t + 1$, $R = t^2 + t + 1$. Then $\gcd(hR, 210) = \gcd(h, R) = 1$. The integer $r_0 = (t + 2R)/4$ satisfies $16tr_0^2 \equiv 1 \pmod{R}$, since $t^3 \equiv 1 \pmod{R}$. Solve $r \equiv r_0 \pmod{R}$ and $r \equiv ((R+1)/4)h^{-1} \pmod{210}$. The progression $p = 4hr - R$ is reduced modulo $840hR$ and is 1 modulo 840. Dirichlet's theorem [2, Theorem 18.1] therefore gives infinitely many prime members. They have the original E state $(a, u) = (hr, hr^2)$, $\alpha = -4t$, and the unavailable canonical word but available sharp word (3). For example the hard prime 825241 has

$$(a, R, u, h, r, s, \kappa) = (206321, 43, 2240439739, 19, 10859, 1, 208402140).$$

At $j = 5$, only $W = 25$ in $\{1, 5, 25\}$ has residue six modulo 19. It returns

$$(a_M, R_M, U, h_M, r_M, s_M, \lambda_M) = (217170, 43439, 25, 1, 5, 43434, 1).$$

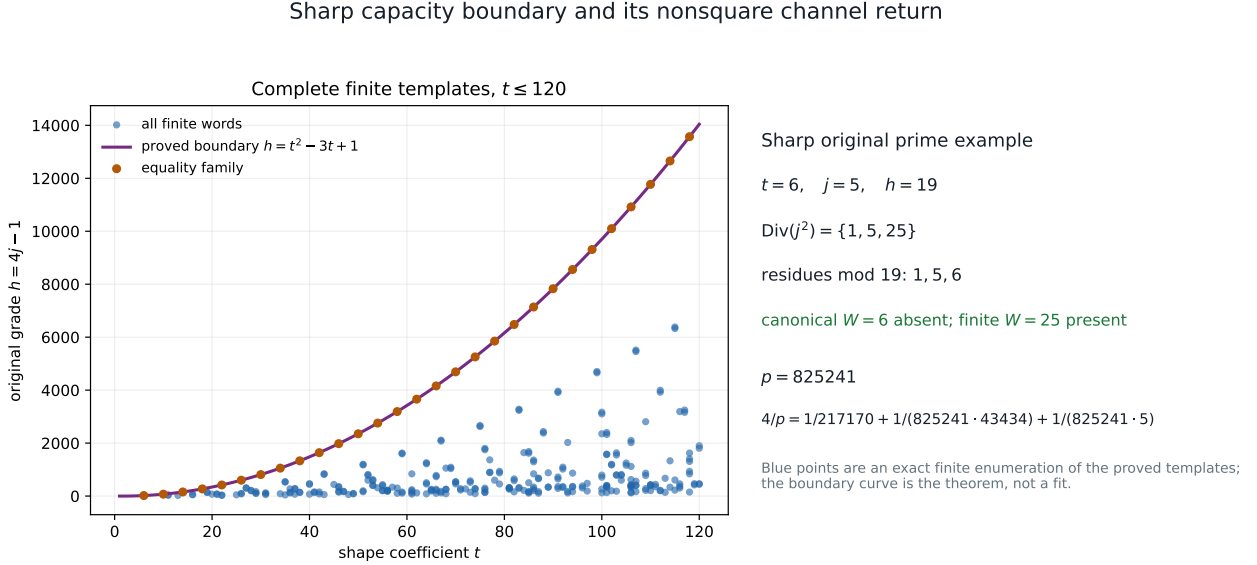


Figure 2: The sharp finite boundary and the exact $p = 825241$ nonsquare return. Blue points enumerate every proved finite template through $t = 120$; the displayed quadratic curve is the theorem.

A complete limitation of same-grade repair. For every fixed $t \geq 4$ there are infinitely many actual hard-prime E states with $\alpha = -4t$ and $\mathcal{A}_t(j) = \emptyset$. Here is the constructive proof. Put $K = \prod_{l \in \mathbb{N}} l^{\lfloor e/2 \rfloor}$ over every prime, including two, and $M = 4K$. For odd A coprime to t , let $\chi_t(A) = (t/A)$ be the Kronecker–Jacobi character determined by the squarefree part of t . The unit classes $A \equiv 3 \pmod{4}$, $\chi_t(A) = 1$ number $\varphi(M)/2$ if t is a square and $\varphi(M)/4$ otherwise. The latter count follows by CRT: the positive quadratic character of t is nontrivial on units 1 modulo four and has value one at -1 . This class set is the singleton $\{-1\}$ only at $t = 1, 2, 3$. Choose $A \neq -1$ and lift it so that $R, R - 4t$ are units at 3, 5, 7. The added restrictions each exclude at most two residues, and are already satisfied when that prime divides K . Dirichlet supplies a prime R in this reduced class with $h = R - 4t > \max(4t, t^2 - 3t + 1)$. Choose $q^2 \equiv t \pmod{R}$, put $r_0 \equiv (4q)^{-1} \pmod{R}$, and impose the same CRT on r as above. Again $p = 4hr - R$ is a reduced progression of primes 1 (mod 840). Its original E state is (hr, hr^2) . But $h \not\equiv -1 \pmod{4K}$ makes $t \nmid j^2$, and Theorem 1 excludes all other words at this grade. Every such prime already has its constructed E solution; this is not an ES counterexample.

For $t = 4$, take $R = 59, h = 43, j = 11$ and $p = 1447321 + 2131080n$. The prime member $n = 3$ is 7840561. The entire capacity set tests 1, 11, 121 modulo 43, giving 1, 11, 35 rather than four. The beta automatic list cannot be extended indiscriminately: actual hard-prime beta sources with coefficients 5, 8, 27 also have empty old-grade boxes; full certificates are included.

Scope and exact records. The full note gives every inverse-source fibre, including its original exterior gate, and corrects a missing gate in the predecessor's prose (its executable already tested it). It preserves every word and ordered denominator. The standard-library programs compare literal divisor boxes and independently constructed primitive source states. Their declared bounds and execution receipts distinguish finite replay from the unbounded proofs above. No initial ES state has been forced at an arbitrary prime; universal occupancy, historical priority, a Lean build, and independent mathematical review are not claimed.

References

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Aust. Math. Soc.*, 94, 50–105. [arXiv:1107.1010](#), §2.
- [2] Sutherland, A. V. (2017). *18.785 Number Theory I, Lecture 18*, MIT, Theorem 18.1.